

Quantifying the Effect of Burn-In Thermal Control on Delivered Acceleration Factor and On-Chip Slack-Sensor Fidelity in an FPGA Target

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Abstract—Burn-in standards such as JEDEC JESD22-A108 and MIL-STD-883 Method 1015 specify apparatus-level thermal conditions for device qualification and inter-laboratory reproducibility; sensor fidelity under those conditions has not been characterized empirically in the FPGA aging-sensor literature. This paper compares two thermal-control architectures — a thermostatic bang-bang oven and a closed-loop PID oven — commanded to the same nominal configuration (80 °C oven-air setpoint, 1.3 V supply) for 23.08 h on the same FPGA device and the same on-chip slack sensor, targeting a 115 °C DUT die-temperature stress condition. Three quantitative findings are reported: (i) the mean delivered junction temperature differs by 9.31 °C across architectures, an Arrhenius acceleration-factor ratio of $1.65\times$ at $E_a=0.7$ eV or 711 h vs. 1182 h of equivalent stress at $T_{\text{use}}=55$ °C; (ii) instantaneous DUT-rail voltage variance is $\sim 1.7\times$ larger under bang-bang even with active software compensation; (iii) instantaneous temperature and voltage jointly explain 92.1% of slack variance under bang-bang but only 59.2% under PID. Together the findings establish that burn-in thermal-control architecture changes both the stress delivered to the DUT and the reversible environmental component superimposed on the sensor output, with direct implications for on-chip sensors consumed by runtime SLM functions such as adaptive voltage scaling and remaining-useful-life estimation.

Index Terms—FPGA aging, burn-in, on-chip aging sensor, thermal control, acceleration factor, JEDEC, MIL-STD.

I. INTRODUCTION

Burn-in is used both to qualify parts and to validate the on-chip aging sensors that increasingly drive Silicon Lifecycle Management (SLM) decisions [1]–[3]. The standards governing burn-in — JEDEC JESD22-A108 [4] and MIL-STD-883 Method 1015 [5] — specify apparatus-level stress conditions aimed at device qualification and reproducibility across laboratories [4]–[6]; the sensor-fidelity behavior of an instrument measuring the DUT during such a test is not part of their declared scope.

On-chip aging sensors infer degradation associated with wear-out mechanisms such as Negative Bias Temperature Instability (NBTI) and Hot Carrier Injection (HCI) from a slack or delay measurement that is reversibly modulated by instantaneous process, voltage, and temperature (PVT). This coupling is well known qualitatively [7], [8], and aging sensors have been validated against standardized burn-in in prior work [9]–[11]. What has not previously been reported is a side-by-side empirical measurement of how the *thermal-control architecture* of the burn-in bench itself affects both the delivered stress and the PVT variance seen by the sensor.

Two burn-in environments are commanded to the same nominal configuration (80 °C oven-air setpoint and 1.3 V supply, targeting a 115 °C DUT die-temperature stress condition) for 23.08 h on the same Artix-7 FPGA, using an auto-tuning slack sensor referenced in [12] unmodified as the measuring instrument. The environments differ only in thermal-control architecture (bang-bang vs. PID); the same-device design isolates measurement contamination introduced by the test environment from device-to-device variability. The findings matter for on-chip sensors later consumed by runtime SLM policies such as adaptive voltage scaling (AVS), remaining-useful-life estimation, and predictive maintenance [1]–[3], [13].

A. Related work

On-chip FPGA aging sensors have been validated against standardized burn-in for over a decade: Valdés-Peña *et al.* [9] validate a configurable aging sensor against MIL-STD-883 Method 1015.50; follow-on work [8], [14] extends the sensor family; Huang *et al.* [10] apply machine-learning correction of temperature-induced measurement error on 28 nm Artix-7 devices; Lanzieri *et al.* [11] survey the landscape of aging detection and monitoring techniques across FPGAs and other embedded-system components; and Bidmeshki *et al.* [7] show in simulation that digital sensors produce false aging alarms when PVT is not explicitly corrected. None performs a side-by-side empirical comparison of two thermal-control architectures on the same DUT and the same sensor, which is what links the physical Arrhenius AF bias to the PVT variance seen by the sensor and is the contribution of this paper.

II. BACKGROUND

A. Arrhenius acceleration

Negative/Positive Bias Temperature Instability and Hot Carrier Injection shift transistor threshold voltages and thus path delay [8], [11], [14], [15]. Burn-in accelerates these mechanisms via the Arrhenius relation

$$AF = \exp \left[\frac{E_a}{k_B} \left(\frac{1}{T_{\text{use}}} - \frac{1}{T_{\text{stress}}} \right) \right], \quad (1)$$

with $E_a \approx 0.7$ eV for silicon-interface mechanisms [4]. For a time-varying stress $T(t)$ the delivered acceleration is $\langle AF \rangle_\tau$, where $\langle AF \rangle$ is the mean of $AF(T(t))$ over the test window. The relevant figure of merit is therefore the trajectory-integrated AF, not the commanded setpoint or the peak temperature.

B. Slack decomposition

An on-chip slack sensor [8], [12] measures total propagation delay of a monitored critical path, which decomposes as

$$t_{\text{slack}}(t, T, V) = t_{\text{slack,aged}}(t) - \Delta t_{\text{PVT}}(T, V), \quad (2)$$

with a permanent, trap-mediated aging term and a reversible PVT term. This decomposition is usually stated as a caveat; this paper quantifies its terms empirically (Sec. IV-C).

C. Standards Context for This Study

JEDEC JESD22-A108 and MIL-STD-883 Method 1015 provide the general experimental context for accelerated burn-in by framing it as a controlled combination of thermal and electrical stress applied under repeatable bench conditions [4], [5]. In that sense, the present study follows the same general structure: both runs are executed on the same DUT, with the same nominal oven-air setpoint, the same supply setpoint, the same sensing infrastructure, and the same supervisory acquisition flow, so that the effect of the thermal-control architecture can be isolated from changes in the rest of the test environment.

For the purposes of this paper, the most relevant standards-related point is that burn-in quality depends not only on the commanded stress configuration, but also on how that configuration is delivered to the powered DUT over time. Accordingly, the analysis focuses on the quantities that directly determine accelerated stress and sensor interpretability in this setup: the delivered die-temperature trajectory, its mean and temporal variation, the corresponding Arrhenius-equivalent stress, and the reversible PVT component superimposed on the sensor output.

III. METHODOLOGY

A. Apparatus and dataset provenance

The DUT is a Xilinx Artix-7 FPGA on a Nexys 4 DDR board [16], instrumented with the auto-tuning slack sensor of [12] used unmodified. The sensor uses MMCM dynamic phase-shift [17] to advance a phase-shifted clock until an XOR comparator fires; each 19.85 ps increment reports one unit of slack. Die temperature and V_{CCINT} are read from the XADC [18]. A programmable PSU supplies core voltage, and a supervisory PC synchronizes all streams at 1 Hz. The two burn-in runs compared in this paper share the same physical DUT and bitstream but were conducted in separate campaigns; between runs the device was power-cycled and the sensor's auto-calibration was re-executed. The analysis therefore relies exclusively on within-run slopes, variance decompositions, and residuals rather than on absolute slack values.

B. Two thermal-control architectures

Two burn-in runs are compared under the same nominal configuration and stress duration, differing only in thermal-control architecture. The nominal configuration comprises an oven-air setpoint of 80 °C and a supply setpoint of 1.3 V, chosen *a priori* to drive the DUT to a 115 °C die-temperature burn-in stress condition under its load; whether either architecture

actually delivers the 115 °C die target is reported in Sec. IV-A. In the *bang-bang* run the oven uses its original thermostatic controller, with a rate-limited software loop (5 mV step, 10-count V_{CCINT} deadband, ≥ 2 s between corrections) added to hold the *mean* V_{DD} at the commanded value; its bandwidth is below the thermostat cycle frequency, so it nulls the long-term mean but not the cycle-scale oscillation. In the *PID* run the same oven is driven by a closed-loop PID controller ($K_p=2.78$, $K_i=1.06 \times 10^{-3}$ /s, $K_d=5.0$ s) obtained by FOPDT identification and SIMC tuning [19], [20]; no V -compensation is needed because the correlated V variation observed under bang-bang does not appear under closed-loop thermal control. Because the same XADC measures die temperature in both runs, any fixed absolute-accuracy offset cancels in the within-run differentials and run-mean differences reported in Sec. IV.

C. Analysis methods

Both runs are 23.08 h long. The first 1.0 h of each run is excluded from steady-state statistics to remove the DUT-package thermal-soak transient. Variance decomposition of the slack signal uses ordinary linear regression; the reported values are ordinary coefficients of determination (R^2) for univariate and multivariate linear models fit to the full post-equilibration window. PVT-corrected residuals are computed by subtracting the fit of slack on T and V jointly. Residual-slope estimates in the three disjoint windows 1–6 h, 6–12 h, and 12–22.5 h use ordinary least squares; confidence intervals are heteroscedasticity-and-autocorrelation-consistent (Newey–West) with lag truncation $L=\lceil 4(n/100)^{2/9} \rceil$ to account for the serial correlation intrinsic to the 1 Hz time series. The thermal-soak time constant is estimated by fitting $A \exp(-t/\tau) + kt + c$ to the PID residual over [1, 22.5] h with non-linear least squares.

IV. RESULTS

A. Delivered temperature and acceleration factor

The two runs reach markedly different steady-state mean DUT die temperatures despite being commanded to the same nominal configuration (Table II, Fig. 1): the bang-bang environment settles at $\bar{T}_{\text{DUT}}=107.73$ °C while PID holds 117.04 °C, a gap of 9.31 °C against the 115 °C die target. In the apparatus used here, the bang-bang oven essentially reached its commanded 80 °C chamber setpoint (the thermostat cut heating near 78 °C within its hysteresis window); the shortfall at the die therefore reflects how thermostatic cycling couples to DUT self-heating. During the off phase the die sheds heat into cooler chamber air, whereas under PID the chamber air is held continuously at 80 °C and the die's self-heating is free to add its full offset on top. Applying Eq. (1) with $E_a=0.7$ eV and $T_{\text{use}}=55$ °C yields $\langle AF \rangle=31.1$ for bang-bang and 51.3 for PID — a delivered-AF ratio of 1.65 $\times$, equivalent to 711 h vs. 1182 h of stress at 55 °C over the same wall-clock interval. Across $E_a \in [0.5, 0.9]$ eV the ratio spans 1.44 $\times$ –1.92 $\times$, so the direction and order of magnitude of the bias do not hinge on the specific E_a . Table I extends this projection across the

TABLE I

ARRHENIUS-EQUIVALENT FIELD-OPERATING TIME DELIVERED BY THE TWO ARCHITECTURES OVER THE SAME 23.08 H WALL-CLOCK BURN-IN ($E_a=0.7$ eV). THE PID/BANG-BANG RATIO IS T_{USE} -INDEPENDENT AND EQUALS $1.65 \times$ AT EVERY ROW.

T_{use} (°C)	Eq. hours, bang-bang	Eq. hours, PID	Eq. days, bang-bang	Eq. days, PID
25	8661	14294	361	596
40	2349	3876	98	162
55	711	1182	30	49
70	243	401	10	17
85	90	149	4	6

TABLE II

STRESS-DELIVERY STATISTICS OVER 23.08 H. MEAN DIE-TARGET ERROR IS COMPUTED AGAINST THE 115 °C DUT DIE-TEMPERATURE TARGET THAT DROVE THE NOMINAL CONFIGURATION.

Metric	Bang-bang	PID
Mean T_{DUT} (°C)	107.73	117.04
$\sigma(T_{\text{DUT}})$ (°C)	2.64	1.25
Mean die-target error (°C)	-9.31	~ 0
$\langle AF \rangle$, $E_a=0.7$ eV	31.1	51.3
AF ratio (PID / bang-bang)	1.65 $\times$	
Mean V_{DD} (V)	1.2988	1.2967
$\sigma(V_{DD})$ (mV)	6.1	3.6

FPGA deployment temperature range: at $T_{\text{use}}=25$ °C, the same wall-clock stress corresponds to ≈ 1.0 year of equivalent field operation under bang-bang and ≈ 1.6 years under PID, a difference a downstream SLM reliability calculation would consume directly.

B. Correlated temperature and voltage variation

In the bang-bang run the instantaneous DUT supply voltage is correlated with the oven oscillation, producing observable V_{DD} drift synchronous with thermal cycles. The software compensation layer nulled the mean V_{DD} (1.2988 V vs. 1.2967 V across the two runs, so the AF result of Sec. IV-A is not confounded by a mean voltage bias) but did not suppress the instantaneous oscillation ($\sigma(V_{DD})=6.1$ mV in bang-bang vs. 3.6 mV in PID; Table II). Because NBTI is commonly modeled with a super-linear dependence on V_{DD} [15], thermal oscillation that correlates with voltage drift at the DUT has a compounding effect on instantaneous stress.

C. PVT-dominated slack variance

The sensor's raw slack trace was regressed on instantaneous T and V by the ordinary-least-squares procedure described in Sec. III-C; Table III reports the resulting R^2 values. In the bang-bang run, temperature alone explains 92.10% of slack variance and jointly with V reaches 92.11%: the large thermal excursion dominates the raw output, and V adds only a marginal contribution once T is known. In the PID run, T and V jointly explain 59.23% of the slack variance — a lower R^2 because the controlled bench presents substantially smaller reversible PVT excursions, leaving less environmentally induced variance to explain. The more relevant post-correction measure is the residual standard deviation after joint regression, which is lower under PID (0.73 cnt) than under bang-bang (0.90 cnt),

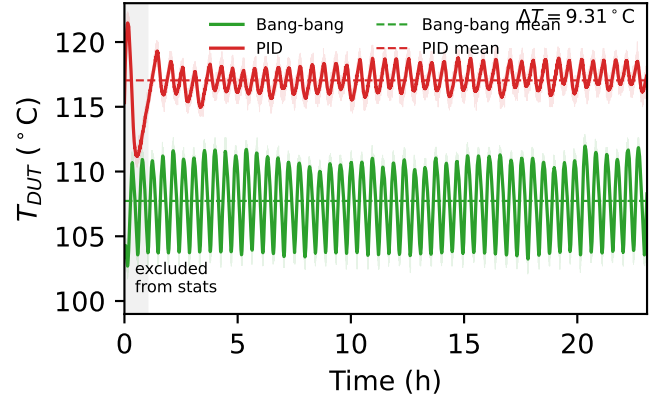


Fig. 1. DUT temperature over 23.08 h for both thermal-control architectures. Dashed lines show run means; the two runs differ in mean by $\Delta T=9.31$ °C, producing the $1.65 \times$ AF ratio of Sec. IV-A.

TABLE III

VARIANCE DECOMPOSITION OF THE RAW SLACK TRACE (R^2 OF ORDINARY LINEAR REGRESSION).

R^2 of linear fit on slack	Bang-bang	PID
T alone	92.10%	54.50%
V alone	60.27%	56.33%
T and V jointly (linear)	92.11%	59.23%
Residual σ after joint regression (cnt)	0.904	0.726

both well below one sensor LSB (19.85 ps) over any sub-hour window.

A windowed analysis of the PVT-corrected residual over three disjoint intervals (1–6 h, 6–12 h, 12–22.5 h, Newey–West 95% CIs) yields statistically overlapping late-window slopes of -6.5 m-cnt/h (bang-bang, $[-11.8, -1.2]$) and -5.3 m-cnt/h (PID, $[-7.7, -2.8]$); over 23.08 h the architectural difference is located in the reversible PVT component and in delivered Arrhenius stress rather than in a separable steady-state residual slope.

V. DISCUSSION AND CONCLUSION

The same FPGA DUT and same on-chip slack sensor, compared under two thermal-control architectures at the same commanded configuration, reveal three consistent consequences: the bang-bang bench delivers substantially lower mean die temperature (and with it a $1.65 \times$ lower Arrhenius AF, 711 h vs. 1182 h of equivalent stress at 55 °C); it imposes larger instantaneous V_{DD} oscillation even with software compensation active; and its sensor output carries a larger reversible PVT component, leaving a 0.90 cnt ≈ 17.9 ps post-correction residual against 0.73 cnt ≈ 14.5 ps under PID — a $\sim 23\%$ wider effective actuation floor on a bang-bang-validated bench. The lower joint R^2 under PID should not be read as worse sensor fidelity but as the signature of a cleaner validation environment. For on-chip sensors later consumed by runtime SLM functions such as AVS, remaining-useful-life estimation, or predictive maintenance, a bench that under-

TABLE IV
ILLUSTRATIVE MEAN SETPOINT-TRACKING BUDGET δ FROM EQ. (3) AT
 $T_{\text{setpoint}}=115^\circ\text{C}$, $E_a=0.7\text{eV}$.

AF target $\pm\eta$	Required δ ($^\circ\text{C}$)
$\pm 5\%$	≈ 0.91
$\pm 10\%$	≈ 1.77
$\pm 20\%$	≈ 3.38

delivers stress while injecting more reversible PVT variance weakens downstream confidence in the resulting signal.

Because the study uses a single DUT over 23.08h, it does not establish a population-level aging distribution and does not resolve a permanent-aging-rate difference between the two architectures: after joint T - V correction, the late-window residual slopes are statistically overlapping, sufficient to quantify reversible contamination but not to separate a distinct permanent-aging trajectory at this duration. The Arrhenius sensitivity expression below is therefore an interpretation of the measured temperature mismatch, not a new aging model.

Because AF depends exponentially on delivered temperature, a first-order expansion of Eq. (1) around T_{setpoint} that requires the AF deviation to remain within a fraction η of nominal yields

$$\delta \approx \frac{k_B T_{\text{setpoint}}^2}{E_a} \ln(1 + \eta), \quad (3)$$

with T_{setpoint} in kelvin. Table IV evaluates Eq. (3) at $T_{\text{setpoint}}=115^\circ\text{C}$ and $E_a=0.7\text{eV}$, converting a mean-temperature-tracking budget into an AF-uncertainty target for a chosen $(T_{\text{setpoint}}, E_a)$ pair. Under that mapping, the measured 9.31°C gap is not a bench artifact but a physically consequential difference in delivered stress.

These findings point to whether burn-in as specified by the JEDEC JESD22-A108 and the MIL-STD-883 Method 1015 is sufficient to validate an FPGA aging sensor. The bang-bang bench misses the apparatus-level chamber condition with its hysteresis window, letting only the PID run satisfy the minimum-ambient clause on which MIL-STD-883 §3.1.1 is based. Beyond that, and under the same nominal configuration, the two benches deliver DUT die temperatures that differ by 9.31°C in the mean and supply-rail variance that differs by $1.7\times$. The standards therefore behave as advertised as qualification specifications: they flag the non-compliant bench and bound chamber-level conditions. What they do not bound is the stress and PVT environment actually experienced by a powered DUT, which is the quantity a critical-path aging sensor directly samples. For sensor-instrumented validation, the use case implicit in SLM workflows built on on-chip aging sensors, that gap is the reporting surface that Eq. (3) and Table IV make concrete.

Future work extends the same protocol to longer campaigns that can resolve permanent aging above the residual floor, to multi-device studies that separate bench effects from process variation, and to multi-temperature experiments that permit direct extraction of E_a . Within the scope of the present study,

thermal-control architecture is a first-order determinant of both delivered stress and measurement interpretability for sensor-instrumented burn-in.

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